EARTHPRESS · PRODUCT REQUIREMENTS AND SPECIFICATIONS

EarthPress product requirements and specifications

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-30	Amish Chadha	PR-007 total set at 205 kg (decision D2, Amish 2026-10-30, EPR-DDR-005): met on paper, margin 1.2 kg
0.1	2026-10-25	Amish Chadha	Kit 1.8 migration of EPR-REQ-001 v0.8 (R1 to R14 become PR-001 to PR-014, old number in "Was"); PR-015 and PR-016 added; traced to EPR-URS-001; status from EPR-CAL-001 v0.7 (PR-007 not met at 203.8 kg; PR-013 USD 542; PR-016 met after fix F1)

EarthPress product requirements and specifications

What the press must do, each requirement traced to a user need in EPR-URS-001 (links in EPR-RTM-001). This document replaces EPR-REQ-001 (v0.8 of 2026-10-02): every requirement keeps its target, verification and status, with its old number in "Was". Status is from the calculation note EPR-CAL-001 v0.7 for the current design (option A, the lever and toggle press). On paper 8 of the 16 requirements are met, two are at risk, one is **not met** (PR-007, mass), one is over its value-engineering target (PR-013, cost) and four can only be verified at TRL 4. Targets remain proposals for review with a co-design partner (EPR-PRB-001). Method: STANDARDS section 22.

The **reference block** is 290 x 140 x 90 mm, about 3.65 L and 6.94 kg dry, from a sandy clay loam stabilized with 5 % cement by dry mass (decided by Amish, 2026-09-25; EPR-DDR-001 items 3 to 5, EPR-DDR-002).

Table 1. Product requirements and specifications. Status from EPR-CAL-001 v0.7 (2026-10-25).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Block dimensions	Blocks shall be 290 x 140 mm on plan and 90 mm high, checked with the block gauge	290 x 140; 90	±2 on plan; ±3 on height	mm	Analysis (CAL-001 sections 5 and 6), TRL 3; gauge checks on a production run, TRL 4	Met on paper: 90.0 mm at the crank stop with the fill within 0 to +2.7 %; mold bulge 0.08 mm	R1
PR-002	UR-002	Performance	Compaction pressure	The press shall reach 2.0 MPa or more on the block face at the end of the stroke	2.0	min	MPa	Analysis (CAL-001 sections 3 to 5), TRL 3; load cell, TRL 4	**At risk**: 2.00 MPa with two operators and a weighed fill; one operator at 500 N stalls at 0.53 MPa; rests on the assumed soil law (STM S6)	R2
PR-003	UR-002, UR-016	Performance	Block strength with suitable soil	Blocks from suitable soil shall reach 4 MPa dry and 2 MPa wet at 28 days (Auroville class B), and in all cases 2.1 MPa (300 psi) as in 14.7.4 NMAC	4; 2; 2.1	min	MPa	Literature, TRL 3; block tests, TRL 4	To verify at TRL 4	R3
PR-004	UR-003	Performance	Output	A crew of four shall make 300 or more blocks in an 8 h day, including mixing	300	min	blocks a day	Analysis (CAL-001 section 9), TRL 3; timed trials, TRL 4	**At risk**: 296 a day on an assumed 85 s cycle (shortfall accepted by Amish, 2026-09-25)	R4
PR-005	UR-004	Usability	Operator force and grip height	The peak pull per operator at the grip shall be 500 N or less, with the grip between	500; 0.8 to 1.9; 1.0	max; band; min	N; m; m	Analysis (CAL-001 sections 3, 4 and 4b), TRL 3;	Met on paper with two operators: 334 N each (base soil), 405 N each	R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				0.8 and 1.9 m above the ground and the peak at 1.0 m or higher				force at the grip, TRL 4	(softer design soil); grip 0.86 to 1.89 m; peak at 1.16 m; eject 469 N for one	
PR-006	UR-005	Physical architecture	Garage-buildable	The press shall be built with a stick welder, angle grinder, drill press and hand tools; turned parts only as bought pins, shafts and bushes	0	max	machined parts made in the shop	Inspection of the model, drawings and build plan (BLD-001), TRL 3	Met by design	R6
PR-007	UR-006	Packaging and transport	Movable	The complete press, including the guard, bolts and bushes, shall weigh 205 kg or less, counting bushes, nuts, washers and weld metal; no piece over 50 kg after removing pins and bolts; it shall fit a 1.5 m pickup bed	205; 50; 1.5	max	kg; kg; m	Analysis (CAL-001 section 8, kit 1.9.1 model), TRL 3; weighing, TRL 4	**Met on paper**: 203.8 kg with every bush, nut, washer and weld bead, 1.2 kg under the 205 kg total (decision D2, decided by Amish 2026-10-30, EPR-DDR-005); heaviest piece 44.5 kg (met); plan 1.28 x 0.58 m (met)	R7
PR-008	UR-007, UR-015	Materials	Durability	The press shall make 50,000 blocks before mold liners or bushes need replacing; all pins in replaceable bushes; a	50,000	min	blocks	Analysis (CAL-001 section 6, fatigue), TRL 3; endurance log, TRL 4	To verify at TRL 4: welded range 89 MPa against 243 MPa, pin range 234 MPa against about 350 MPa; wear	R8

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				grease point on every pivot					needs soil data	
PR-009	UR-008	Usability	Soil test kit	Field tests (jar sedimentation, shrinkage box, ribbon and drop tests) shall give a go or no-go and a starting cement content, a first screen within 1 h and a full result within 24 h	1; 24	max	h	Inspection of the written procedure (BLD-001), TRL 3; comparison with laboratory grading, TRL 4	Met by design; accuracy to verify at TRL 4	R9
PR-010	UR-010	Materials	Low stabilizer use	Blocks shall reach PR-003 with 8 % cement by dry mass or less; 5 % is the design value	8	max	% of dry mass	Literature, TRL 3; block tests, TRL 4	To verify at TRL 4 (design at 5 %, 0.33 kg a block)	R10
PR-011	UR-011	Materials	Embodied carbon	Cradle-to-gate CO2 of the walling material shall be 25 % or less of a fired-brick wall of equal volume	25	max	%	Analysis (CAL-001 section 10, cited factors), TRL 3	Met on paper: 7.6 % (0.35 t against 4.55 t for a small house)	R11
PR-012	UR-012	Safety	Safe operation	The lever shall not fall freely when released under load; the lid latch shall hold at least 1.5 times the peak block force; pinch points between lever and frame shall be guarded or have 100 mm	1.5; 100	min	x 81.2 kN; mm	Analysis and design review (CAL-001 section 6), TRL 3; proof-load test, TRL 4	Met on paper: rest catch and end pawl; lid and latch rated 173 kN; linkage guard	R12

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				or more clearance						
PR-013	UR-009	Cost	Affordable	Press and soil kit parts shall cost at or below the value-engineering target, excluding labor and cement	500	max	USD	Analysis of the priced BOM (`bom/bom.csv`), TRL 3	**Over target**: USD 542 (indicative), USD 42 over; savings in EPR-DEC-001	R13
PR-014	UR-013	Usability	Open and documented	CERN-OHL-S drawings, BOM and build plan shall let a welder build the press without contacting the author	1	min	outside build review	Review by an outside fabricator, TRL 4	To verify at TRL 4	R14
PR-015	UR-014	Usability	Human powered	The press shall need no electricity, fuel or hydraulic fluid to press or eject a block	0	max	powered parts	Inspection of the model and BOM, TRL 3	Met by design	
PR-016	UR-012, UR-007	Safety	Structure at the design load	Every member in the load path shall stay below yield at the 173 kN design load (1,000 N at the grip at the crank stop)	1.0	min	safety factor on yield	Analysis (CAL-001 sections 6 and 6b), TRL 3; proof-load test, TRL 4	Met on paper after fix F1 (EPR-DEC-001): lowest factor 1.29; the 35 mm C45 pins did not meet it	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Requirements not met or at risk

- **PR-007 now met on paper, with a thin margin:** 203.8 kg against the 205 kg total decided on 2026-10-30 (EPR-DDR-005); the earlier 199.996 kg left out the bushes, nuts, washers and weld metal, and fix F1 adds about 1.2 kg. Any growth above 1.2 kg reopens it.

- **PR-013 over target:** USD 542 against the USD 500 value-engineering target (a hypothetical control target, `budget_usd` unchanged).
- **PR-002 at risk:** met only with two operators, a weighed fill and the assumed soil stiffness.
- **PR-004 at risk:** 296 against 300 blocks a day on an assumed cycle; the shortfall was accepted by Amish on 2026-09-25 (EPR-DDR-002).
- PR-003, PR-008, PR-010 and PR-014 can only be verified at TRL 4, which is on hold.

Revisions accepted (EPR-DDR-002)

Decided by Amish, 2026-09-25: go with recommendation. R5 (now PR-005) was a peak pull of 500 N or less with the grip between 0.9 and 1.6 m; R7 (PR-007) was 140 kg, then 190 kg, and was 200 kg including the guard and bolts from 2026-10-02 (EPR-DDR-003 A1) and has been 205 kg, counting bushes, nuts, washers and weld metal, since 2026-10-30 (EPR-DDR-005); R13 (PR-013) was USD 450 and has been USD 500 since 2026-09-25.

Assumptions

- Compaction pressure of 2 to 4 MPa and about 5 % cement follow Auroville Earth Institute practice (Auroville Earth Institute).
- Dry block density about 1,900 kg/m³ (estimate), giving about 6.9 kg per dry block.
- One operator can pull 500 N at the grip for short strokes, using body weight; two can pull 1,000 N on the T-handle. To be checked with users.
- Soil pressure rises tenfold over the last 14 mm of the stroke (range 10 to 20 mm); to be measured with partner soils (EPR-STM-001 S6).
- A crew of four covers soil sieving, mixing, pressing and stacking.